Probing-Resistant Masked Hardware: Formal Security Proofs and Simulation-Based Side-Channel Verification

Cheonshik Kim

***Abstract*—Masking is the prevailing algorithmic countermeasure against power and electromagnetic side-channel analysis of cryptographic hardware, yet a persistent gap remains between the abstract security order claimed by a masking scheme and the leakage observed once physical effects such as glitches and transitions are taken into account. Our central contribution is to *characterize* this soundness gap in equipment-free evaluation: a value-based statistical verifier can accept a design that a glitch-extended probe deterministically breaks. The model-level distinction between value-based and glitch-extended probes is classical [1]; what has been missing is a characterization of *when* the cheaper channel's verdict is sound. We give such a characterization via a cone-non-completeness condition on register-input cones, prove that passing the corresponding sound-by-construction lint forces the value-based and glitch-extended verdicts to coincide, and exhibit an explicit first-order witness that satisfies the value-based verifier while a single robust probe recombines the secret. We implement the check on the flattened gate-level netlist (seconds of run time): it accepts the final first-order design, flags every cone that a gate-level statistical test shows to leak in an earlier variant without operand refresh, and rejects controlled violations (mask removal, register collapse). We pair this static channel with a masked hardware construction whose multiplication gadget is proven robust d-NI (for arbitrary d) via a perfect-simulation argument. On the first-order AES S-box built from it, PROLEAD reports 0 leaking probing sets out of $14{,}400$ at $512{,}000$ simulations, while gate-level power simulation with TVLA reports no detection; the TVLA/MI channel has low detection power on its Hamming-distance traces, so we treat it as supporting evidence only. The design uses 56 fresh-random bits per invocation and a 17-cycle latency, i.e. more randomness and latency than DOM or HPC2 S-boxes; we do not claim a cost advantage. Empirical evaluation is limited to $d{=}1$. Together these results connect a provable gadget-level guarantee with reproducible, equipment-free verification at first order.**

***Index Terms*—Side-channel analysis, masking, threshold implementation, probing security, formal verification, TVLA, hardware security, cryptographic hardware.**

I. Introduction

CRYPTOGRAPHIC hardware such as AES accelerators can leak secret key material through physical side channels: an adversary who measures the instantaneous power consumption or electromagnetic (EM) emanations of a device during encryption can recover the key using differential or correlation power analysis (DPA/CPA) [2], [3]. *Masking* is the most widely deployed algorithmic countermeasure: every secret-dependent value x is split into $d+1$ random shares $x^{(0)}, \ldots, x^{(d)}$ such that any d of them are jointly independent of x, forcing the attacker to combine $d+1$ leakage points and thereby raising the measurement cost exponentially in the noise level [4], [5].

Despite two decades of progress, a fundamental gap persists between the *claimed* security order of a masking scheme and the leakage *actually observed* in silicon. The root cause is that physical hardware violates the idealized assumptions of the abstract model: combinational glitches and signal transitions recombine shares within a single clock cycle, degrading a nominally d-th order secure design to a lower effective order [6], [7]. Threshold Implementations (TI) and their successors (DOM, HPC) address glitches through non-completeness and register barriers [7]–[9], but verifying that a concrete netlist attains its intended order remains error-prone.

Our approach. We argue that the probing-security theory developed for secure multiparty computation (MPC) [4] provides exactly the right abstraction to close this gap when it is (a) lifted to the glitch-extended hardware model and (b) paired with reproducible simulation-based verification. Building on this observation, we design a masked hardware construction and prove its security in the robust d-probing model [1], then check the first-order instance with equipment-free, simulation-based tools.

Contributions. This paper makes the following contributions:

- We characterize the *soundness gap* between equipment-free value-based statistical verification and glitch-extended probing, and give a sound-by-construction cone-non-completeness lint that closes it at first order: passing the lint certifies that the two verdicts coincide at first order, and failing it exhibits a concrete recombination witness (Section V-A, Definition 3, Theorems 3 and 4, Corollary 3).
- We back the gap-witness theorem with a fully worked constructive artifact in the Supplemental Material: an algebraic derivation of the first-order witness, a cone that deterministically recombines the secret under a glitch-extended probe while its registered value remains uniform and independent of the secret, establishing the concrete gap of Theorem 4 (Supplemental Material, Section S2).
- We implement the lint on the flattened synthesized netlist ($67{,}344$ combinational gates, 818 flip-flops; run time in seconds) and cross-check it against a gate-level statistical test on the final design, on an earlier variant without operand refresh, and on two controlled violations (Table I, Fig. 1).
- We give sufficient conditions under which our masked gate construction attains order-d security in the robust (glitch-extended) probing model, and prove them formally (Section V).

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C. Kim is with the Department of Computer Engineering, Sejong University, Seoul, Republic of Korea (e-mail: mipsan@sejong.ac.kr).

- We provide a simulation-based verification methodology combining the formal tool PROLEAD with gate-level power simulation and TVLA, and report that, at first order, the observed results are consistent with the proven order (Sections VI–VII); together with the static lint above, this yields three equipment-free verification channels on the same netlist, whose differing detection powers we characterize explicitly rather than treat as interchangeable.
- On protected AES S-box benchmarks we compare against TI, DOM, and HPC on fresh randomness and latency, and report that our construction is more expensive on both (Table V, Section VII).

The remainder of the paper is organized as follows. Section II reviews masking and probing models. Section III states the threat model and security definitions. Section IV presents our construction. Section V contains the formal security analysis. Sections VI and VII describe the verification methodology and experimental results. Section VIII surveys related work, and Section IX concludes.

II. Background and Preliminaries

A. Boolean Masking

Let $x \in \mathrm{GF}(2^n)$ be a sensitive value. A $(d+1)$-sharing of x is a tuple $(x^{(0)}, \ldots, x^{(d)})$ with $x = \bigoplus_{i=0}^{d} x^{(i)}$, where $x^{(0)}, \ldots, x^{(d-1)}$ are drawn uniformly at random and $x^{(d)}$ is chosen to satisfy the invariant. Linear functions are applied share-wise; the difficulty lies in securely evaluating non-linear operations (e.g., the AES S-box multiplicative inverse in $\mathrm{GF}(2^8)$).

B. Threshold Implementations and Successors

Threshold Implementations [7] enforce three properties — correctness, non-completeness, and uniformity—so that no combinational function depends on all shares of any secret, which provides provable first-order security even in the presence of glitches. Domain-Oriented Masking (DOM) [8] and Hardware Private Circuits (HPC) [9] generalize this to arbitrary orders while reducing fresh randomness and enabling composability.

C. Probing Models

In the d-probing model [4], an adversary places up to d probes on internal wires and learns their exact values; a circuit is secure if any such set of d probes is independent of the secret. The *glitch-extended* (robust) probing model [1] accounts for combinational recombination by letting a single probe on a wire reveal all stable inputs to the combinational cone feeding it since the last register stage. We adopt the robust model throughout.

D. Relation to Prior Verification Frameworks

Several tools already establish or check probing security for masked hardware. We briefly contrast them with the sound-lint proposed in §V-A to make the contribution of this paper precise.

a) SILVER [10] and exact verifiers.: SILVER performs an exact, BDD-based analysis of glitch-extended probes on a gate-level netlist and issues a definitive verdict for the given design; exact procedures of this kind are costly on large netlists, which is why full-S-box and full-round designs are usually checked with composition arguments or statistical tools. Bloem *et al.* [11] established the probing security of masked hardware in the presence of glitches by exact formal verification, and Shahmirzadi *et al.* [12] verified first-order masked AES implementations with SILVER. MaskVerif [13] and IronMask [14] decide probing-style security properties of masked implementations, and VERICA [15] targets combined leakage-and-fault attacks. All of these decide properties of the circuit model itself; none exposes an intermediate abstraction whose verdict can be certified sound with respect to a statistical leakage bound, so their outputs cannot be composed with a DDF14-style noisy-leakage argument without re-running the underlying decision procedure.

b) PROLEAD [16].: PROLEAD is simulation-based: it enumerates glitch-extended probe sets up to a fixed order and runs statistical tests over sampled traces. PROLEAD is highly precise on the traces it observes and is our gold-standard cross-check throughout §VII-C. Its guarantees, however, are asymptotic in the number of simulations and offer no completeness certificate: a probe set that is not sampled cannot be ruled leaky, and a scheme that passes at 5×10^5 simulations may fail at 10^7. PROLEAD therefore answers "does this design appear secure under the simulation budget?" rather than "is every probe cone provably non-leaking?"

c) HPC2 / PINI [9], [17].: HPC2 and the broader PINI methodology are compositional frameworks: they provide gadget-level primitives whose composition is guaranteed to remain secure. They are constructive rather than diagnostic—once a design is expressed as a composition of PINI gadgets, security follows by inspection—but they do not verify arbitrary post-synthesis netlists such as those emitted by Yosys optimization passes, where original gadget boundaries are erased.

d) Cardinal and algebraic composition [18], [19].: Recent algebraic verifiers such as Cardinal and its precursors target random-probing composition and achieve strong asymptotic bounds under the assumption of noisy wires. Their verdicts are essentially algebraic; the required per-wire noise threshold is not directly measured from the target netlist but stipulated as an assumption. Our robust-probing claims are complementary: they are checked on the netlist itself and make no assumption on the measurement noise.

e) This work: sound-lint as a certified pre-filter.: Positioned at a strictly lower cost than SILVER and PROLEAD —linear-time union-find on the flattened netlist (§V-B, seconds on $67{,}344$ gates)—our lint does not attempt to be a decision procedure for probing security. Instead, Theorem 3 proves that a PASS verdict rules out an entire structural class of first-order leakage (co-share appearance in a common glitch cone), and Theorem 4 exhibits an equivalence-preserving netlist perturbation on which SILVER and PROLEAD agree but no purely combinational lint can. The lint is therefore a soundness-certified pre-filter that complements the exact but heavier tools rather than replacing them, and its cross-check

against a statistical test (Table I) empirically supports this positioning.

III. Threat Model and Security Definitions

We consider an adversary $\mathcal{A}$ who obtains noisy observations of the power/EM leakage of a masked hardware implementation during operation on a fixed but unknown key. $\mathcal{A}$ may choose plaintexts adaptively and collect an unbounded number of traces. We abstract physical leakage by the robust d-probing model and require security in the following sense.

Definition 1 (Robust d-probing security)**.** *A masked circuit C is* robust d-probing secure *if, for every set $\mathcal{P}$ of at most d glitch-extended probes, the joint distribution of the probed values is independent of the secret x; equivalently, the mutual information $I(x;\mathcal{P}) = 0$.*

Definition 2 (Composability)**.** *A gadget is d-Non-Interfering (NI) if any d probes can be simulated using at most d of its input shares, and d-Strong-Non-Interfering (SNI) if the simulation additionally separates internal and output probes. SNI gadgets compose to yield larger d-probing secure circuits [20].*

IV. Proposed Masking Construction

We target the non-linear core of the AES S-box, i.e., the multiplicative inverse in $\mathrm{GF}(2^8)$, computed with the Itoh–Tsujii addition chain $x^{-1} = x^{254}$: a chain of share-wise Frobenius squarings $(x^2, x^4, \ldots, x^{128})$ interleaved with six field multiplications ($x^2{\cdot}x^4$, $x^6{\cdot}x^8$, $x^{14}{\cdot}x^{16}$, $x^{30}{\cdot}x^{32}$, $x^{62}{\cdot}x^{64}$, $x^{126}{\cdot}x^{128}$). Linear layers (squarings and the final AES affine map) are shared component-wise; the multiplier is the only gadget requiring fresh randomness, apart from one operand refresh detailed below.

Our multiplier gadget builds on the domain-oriented masking (DOM) multiplier [8]. The two secret inputs $a, b \in \mathrm{GF}(2^m)$ are given as independent $(d{+}1)$-sharings $(a^{(i)})_i$ and $(b^{(j)})_j$, with each share assigned to its own *domain*. The gadget produces a $(d{+}1)$-sharing $(c^{(k)})_k$ of $c = a \cdot b$ in three stages: (i) all domain products $p_{ij} = a^{(i)} \cdot b^{(j)}$; (ii) a *cross-domain refresh* in which each unordered pair $\{i, j\}$ with $i \neq j$ draws a single fresh mask $r_{ij} = r_{ji}$ that is added to both p_{ij} and p_{ji}; and (iii) a *register barrier* that stores every refreshed term before the domain compression $c^{(i)} = \bigoplus_j q_{ij}$. Sharing one mask across the symmetric pair guarantees correctness (the masks cancel in $\bigoplus_i c^{(i)}$), while the register barrier blocks glitch propagation across domains. Algorithm 1 states the order-d gadget precisely; it consumes $R(d) = \binom{d+1}{2}$ fresh m-bit mask words ($m{=}8$ here) per multiplication.

a) Operand independence in the Itoh–Tsujii chain.: The independence precondition on the multiplicand sharings is *not* automatic in the addition chain. Squaring is $\mathbb{F}_2$-linear, so it is applied share-wise, and the sharings of x^{2^i} and x^{2^j} produced by the chain remain *aligned* with the input sharing: $(x^{2^k})^{(i)} = \left(x^{(i)}\right)^{2^k}$. Feeding two such aligned sharings directly into Algorithm 1 breaks the gadget's precondition: a glitch-extended probe on the cross-term cone of the first multiplication ($x^2{\cdot}x^4$) observes $\{(x^2)^{(0)}, (x^4)^{(1)}\}$, from which $(x^{(0)})^4 \oplus (x^{(1)})^4 = (x^{(0)} \oplus x^{(1)})^4 = x^4$ is recombined

Algorithm 1 Proposed order-d masked multiplier

Input: independent sharings $(a^{(i)})_{i=0}^{d}$, $(b^{(j)})_{j=0}^{d}$
Output: sharing $(c^{(k)})_{k=0}^{d}$ with $c = a \cdot b$
sample fresh $r_{ij} \in \mathrm{GF}(2^m)$ for all $0 \le i < j \le d$; set $r_{ji} \leftarrow r_{ij}$
for $i = 0$ **to** d **do**
  for $j = 0$ **to** d **do**
    $p_{ij} \leftarrow a^{(i)} \cdot b^{(j)}$ {domain product}
    if $i = j$ **then**
      $q_{ij} \leftarrow \mathrm{Reg}(p_{ij})$
    else
      $q_{ij} \leftarrow \mathrm{Reg}(p_{ij} \oplus r_{ij})$ {refresh + barrier}
    end if
  end for
end for
for $i = 0$ **to** d **do**
  $c^{(i)} \leftarrow \bigoplus_{j=0}^{d} q_{ij}$ {domain compression}
end for
return $(c^{(k)})_{k=0}^{d}$

deterministically by one offline squaring. We therefore insert a single registered refresh on one operand of the first multiplication, $(x^4)'^{(i)} \leftarrow \mathrm{Reg}((x^4)^{(i)} \oplus r_6)$, where one fresh byte $r_6 \in \mathrm{GF}(2^8)$ is added to both shares, $i \in \{0, 1\}$ (the mask cancels in $\bigoplus_i (x^4)'^{(i)}$, so correctness is preserved). Its role is blinding, not joint re-randomization: each refreshed operand register is a blinded value in the sense of Definition 3, so every cross-term cone of the first multiplication carries at most one unblinded share index, and the recombination of Theorem 4 is masked by the fresh pad. Because the refresh does not decorrelate the two multiplicand sharings jointly, the netlist-level claim is made through cone non-completeness (Corollary 1, Theorem 3), which the implemented netlist passes. In every later multiplication one operand is the previous multiplier's output, already re-masked by that stage's r_{01}; every cross-term cone of the next stage then pairs a blinded prior output with a single chain share, which preserves cone non-completeness. The first-order S-box thus consumes 6 refresh bytes r_{01} (one per multiplication) plus 1 operand-refresh byte, i.e. 56 fresh-random bits per invocation in total.

V. Formal Security Analysis

We establish, in order, correctness, output uniformity, glitch-robust non-completeness, robust d-NI security with perfect simulation, first-order composition through cone non-completeness, and the cost characterization. Throughout, $\mathrm{Reg}(\cdot)$ denotes a synchronous register, and every combinational cone is delimited by the register barrier of Algorithm 1.

Lemma 1 (Correctness)**.** *The output of Algorithm 1 satisfies $\bigoplus_{i=0}^{d} c^{(i)} = a \cdot b$.*

Proof. By construction, $\bigoplus_i c^{(i)} = \bigoplus_i \bigoplus_j q_{ij} = \bigoplus_{i,j} p_{ij} \oplus \bigoplus_{i \neq j} r_{ij}$. Each unordered pair $\{i, j\}$ contributes $r_{ij} = r_{ji}$ exactly twice, once in q_{ij} and once in q_{ji}, so $\bigoplus_{i \neq j} r_{ij} = 0$. Hence $\bigoplus_i c^{(i)} = \bigoplus_{i,j} a^{(i)} b^{(j)} = \left(\bigoplus_i a^{(i)}\right)\left(\bigoplus_j b^{(j)}\right) = a \cdot b$. □

Lemma 2 (Output uniformity)**.** *If the input sharings are uniform and independent, then every d of the output shares $(c^{(i)})_{i=0}^{d}$ are jointly uniform and independent of c.*

Proof. Fix any d output shares, i.e., all but $c^{(k)}$. For each $i \neq k$ the term $q_{ik} = \mathrm{Reg}(a^{(i)}b^{(k)} \oplus r_{ik})$ carries the fresh mask $r_{ik} = r_{ki}$, which occurs in $c^{(i)}$ and, among all output shares, only additionally in the excluded $c^{(k)}$. Thus the d masks $\{r_{ik}\}_{i\neq k}$ are mutually independent and uniform, each appearing in exactly one retained share. Conditioned on all remaining quantities, the map from $(r_{ik})_{i\neq k}$ to the d retained shares is a bijection, so they are jointly uniform and independent of c. □

Lemma 3 (Glitch-robust non-completeness)**.** *In the robust (glitch-extended) probing model, the joint view of any single probe inside Algorithm 1 is a function of at most one share of a, at most one share of b, and fresh uniform values that are independent of everything else.*

Proof. The register barrier partitions the circuit into two combinational regions. *(i) Before the barrier.* Each cone computes an input share, a single domain product $p_{ij} = a^{(i)}b^{(j)}$, or its refreshed value $p_{ij} \oplus r_{ij}$. A glitch-extended probe reveals its stable inputs, namely $a^{(i)}$, $b^{(j)}$, and possibly r_{ij}: one share of each secret. No pre-barrier cone combines two distinct a-shares or two distinct b-shares. *(ii) After the barrier.* A probe on any wire of the compression network $c^{(i)} = \bigoplus_j q_{ij}$ reveals (a subset of) the registered values $\{q_{ij}\}_j$, whose glitches were absorbed by $\mathrm{Reg}(\cdot)$. The diagonal value $q_{ii} = a^{(i)}b^{(i)}$ is *unmasked* and depends on exactly one share of each secret. Every off-diagonal value $q_{ij} = p_{ij} \oplus r_{ij}$ $(j \neq i)$ is a one-time pad by a mask r_{ij} that occurs in no other member of the set, so jointly $\{q_{ij}\}_{j\neq i}$ is uniform and independent of q_{ii}. Hence the whole set is distributed as a function of $(a^{(i)}, b^{(i)})$ and independent uniform values, although the individual q_{ij} involve different b-shares. □

Lemma 4 (Joint mask independence)**.** *Consider Algorithm 1 on shares of a, b and fix any set of adversary probes. For an unordered pair $e = \{i, j\}$, $i \neq j$, let $r_e = r_{ij} = r_{ji}$, and let $\mathcal{U}$ be the set of pairs e whose mask r_e is not itself probed. Fix an orientation (i, j) of each $e \in \mathcal{U}$. Conditioned on the shares of a, b and all probed masks, the tuple $(q_{ij})_{e\in\mathcal{U}}$ is uniform on $\mathrm{GF}(2^m)^{|\mathcal{U}|}$ and independent of every probe whose expression does not involve $\{r_e\}_{e\in\mathcal{U}}$. The opposite orientation is determined by $q_{ji} = q_{ij} \oplus p_{ij} \oplus p_{ji}$.*

Proof. Each r_e is drawn uniformly and independently and occurs only in the two registered terms q_{ij} and q_{ji} (and in the compression sums that read them). For $e \in \mathcal{U}$ the mask is unobserved, so $q_{ij} = p_{ij} \oplus r_e$ is a one-time-pad translation of the deterministic value p_{ij} by an independent uniform element; the translations of distinct pairs are by independent masks, hence the tuple is jointly uniform. The opposite orientation shares the same mask, so $q_{ij} \oplus q_{ji} = p_{ij} \oplus p_{ji}$ holds identically and the two orientations must not be treated as independent. Any probe not involving the masks $\{r_e\}_{e\in\mathcal{U}}$ is a function of variables independent of them. □

Theorem 1 (Robust d-NI with perfect simulation)**.** *In the robust d-probing model of Definition 1, Algorithm 1 satisfies:*

(i) Robust d-NI. *For any adversary placing $t = t_1 + t_2 \leq d$ probes (t_1 on internal wires, t_2 on wires of the compression network), there exists a probabilistic simulator that reproduces the joint probe view from at most t shares of a and t shares of b.*

(ii) Perfect simulation. *The simulator's output distribution equals the real joint distribution of the probes exactly (statistical distance zero). Consequently the probes carry no information about a or b beyond the requested shares, in the information-theoretic sense.*

Proof. Probe classification. By Lemma 3, the view of any single probe consists of registered values of three kinds: input-share registers $a^{(i)}$ and $b^{(j)}$, mask registers r_e (one per unordered pair $e = \{i, j\}$, $i \neq j$, with $r_e = r_{ij} = r_{ji}$), and product registers q_{ij}. Call the pair e *observed* if some view contains one of its orientations (q_{ij} or q_{ji}), call its mask *observed* if the register r_e appears in some view, and let $\mathcal{U}$ be the set of observed pairs with unobserved masks (the set $\mathcal{U}$ of Lemma 4). Assign to each probe an index pair: to a pre-barrier cone the pair (i, j) of its share-register stable inputs (either index may be absent); to a probe on any wire of row k of the compression network the pair (k, k), whether or not the observed subset contains the diagonal; to a probe on a mask register no pair. Let I and J collect the first and second coordinates of all assigned pairs. Each probe contributes at most one index to each of I and J, so $|I|, |J| \leq t$, which is the share budget of part (i) once the simulator below is shown to reproduce the view. The simulator requests $\{a^{(i)}\}_{i\in I}$ and $\{b^{(j)}\}_{j\in J}$; the row assignment (k, k) is what a compression probe needs for the unmasked diagonal q_{kk}, and this is why the gadget is d-NI and not d-SNI in the robust model. Every observed value is simulated as follows.

Rule 1 (share and mask registers). An observed share register is copied from the requested shares. An observed mask register r_e is sampled uniformly and independently.

Rule 2 (unmasked diagonal). An observed diagonal q_{kk} is computed as $a^{(k)}b^{(k)}$ from the requested shares; both indices lie in I and J by the row assignment.

Rule 3 (masked orientations). For every $e = \{i, j\} \in \mathcal{U}$ draw one uniform ρ_e. If exactly one orientation of e is observed, set it to ρ_e. If both orientations are observed, set

$$q_{ij} = \rho_e, \qquad q_{ji} = \rho_e \oplus p_{ij} \oplus p_{ji}, \qquad p_{ab} = a^{(a)}b^{(b)},$$

computing p_{ij} and p_{ji} from the requested shares, with Rule 5 applying to any missing factor; when the two orientations are observed by row probes, the assignments (i, i) and (j, j) already place all four factors in $I \times J$. Sharing ρ_e between the two orientations reproduces the deterministic relation $q_{ij} \oplus q_{ji} = p_{ij} \oplus p_{ji}$ of the real view (the coset structure of Lemma 4); sampling the orientations independently would leave this relation unmatched and would give a positive statistical distance.

Rule 4 (observed-mask orientations). For an observed orientation q_{ij} of a pair whose mask is observed, set $q_{ij} = p_{ij} \oplus r_e$

with the r_e of Rule 1, again computing p_{ij} from the requested shares, with Rule 5 applying to missing factors.

Rule 5 (dummy substitution). In Rules 3 and 4, any factor $a^{(i)}$ with $i \notin I$ or $b^{(j)}$ with $j \notin J$ (an *extra* index, in practice only for b) is replaced by an independent uniform value, one dummy per extra index, used consistently wherever that index occurs. This is valid because of the following two facts.

(a) Counting. Extra indices arise only for b, and only in Rule 4: an observed orientation $q_{ij} = a^{(i)}b^{(j)} \oplus r_e$ is observed through a row-i probe, whose pair (i, i) supplies $a^{(i)}$ and $b^{(i)}$ but not $b^{(j)}$. (If both orientations of e are observed, the row assignments (i, i) and (j, j) already place all four factors in $I \times J$; a pre-barrier cone probe carries its own pair and needs no dummy.) An extra index j thus needs the mask register r_e to be observed as well, by some probe π_e other than the row probe. If π_e is a probe on the register r_e itself, it was assigned no pair and contributes nothing to J. If π_e is a pre-barrier cone probe on q_{ij} or q_{ji}, its pair has second coordinate j or i; the first would put j in J, so j would not be extra, and the second is already supplied to J by the row-i probe. In either case we charge the extra index j to π_e, and π_e's own (possible) contribution to J is not needed. Distinct extra indices are charged to distinct probes, and no probe is charged both for an extra index and for a needed element of J. Hence, with m the number of probes charged for extras, $|E_b| \le m$ and $|J| \le t - m$, so $|J \cup E_b| \le t \le d$; also $|I| \le t \le d$.

(b) Uniformity. Any d shares of a $(d+1)$-sharing are jointly uniform and independent of the shared secret, for every value of the secret. By (a), the shares $\{b^{(j)}\}_{j \in J \cup E_b}$ are jointly uniform and independent of b, so conditioned on the requested shares $\{b^{(j)}\}_{j \in J}$ the extra shares are independent uniform values; the a- and b-sharings are independent. Replacing the extra shares by independent uniform dummies therefore leaves the joint law of the whole view unchanged.

Compression wires. A probe on a wire of $c^{(k)} = \bigoplus_j q_{kj}$ observes a subset of $\{q_{kj}\}_j$ together with the probed wire itself, whose value is the XOR of the simulated terms; Rules 1–5 cover everything in its view.

Proof of (i). Rules 1–5 output the complete joint view from at most $|I| \le t$ shares of a and $|J| \le t$ shares of b.

Proof of (ii): perfect simulation. The real view is a deterministic function of (a) the requested shares, (b) the observed masks, and (c) the pair values $(q_{ij})_{e \in \mathcal{U}}$ of the masked orientations. Conditioned on (a) and (b), the tuple (c) is jointly uniform and independent of all remaining observed values by Lemma 4. The simulator draws (a) and (b) from their real distributions and (c) uniformly (Rule 3); every remaining observed value is the same deterministic function of (a)–(c) and of the dummy values in both worlds (Rules 2, 4, 5), and by Rule 5 the dummy values enter only through factors whose conditional law given the rest is uniform. Hence the simulated and real views have identical joint distributions: the statistical distance is exactly zero, and $\mathrm{I}(\mathrm{view}; a, b \mid \{a^{(i)}\}_{i \in I}, \{b^{(j)}\}_{j \in J}) = 0$, so the t probes carry no information about a, b beyond the requested shares. □

Corollary 1 (First-order composition)**.** *Let N be a first-order circuit built from instances of Algorithm 1 and share-wise linear maps, and suppose every register-input cone of N is cone non-complete in the sense of Definition 3. Then N is robust 1-probing secure.*

Proof. This is Theorem 3. Algorithm 1 is the source of the blinded values (Lemma 3), and share-wise linear maps preserve share indices. The only place where aligned sharings of one secret meet in a multiplier is the first multiplication $x^2 \cdot x^4$ of the Itoh–Tsujii chain; there the registered operand refresh (a single fresh byte r_6 added to both shares) of Section IV makes each refreshed operand register a blinded value, which removes the violation exhibited in Theorem 4 without jointly re-randomizing the sharings, and the lint of Section VII confirms that the resulting netlist passes. □

Theorem 2 (Full-chip first-order security of round 1)**.** *Let* masked_aes_round1_first_order *be the design composed of 16 instances of Algorithm 1-based S-boxes interleaved with the share-wise linear maps ShiftRows, MixColumns, and AddRoundKey, with independently shared state bytes and an independently shared round key. If every register-input cone of its netlist is cone non-complete (Definition 3), it is robust 1-probing secure.*

Proof. By Theorem 3 it suffices to bound what a single glitch-extended probe reveals, i.e. the fan-in of one register-input cone. *Cones inside an S-box* are those of Corollary 1. *Cones in the linear layers.* ShiftRows permutes bytes within each share; MixColumns is $\mathbb{F}_{2^8}$-linear and is applied to each share separately, $y^{(i)} = M \cdot x^{(i)}$; AddRoundKey computes $y^{(i)} = x^{(i)} \oplus k^{(i)}$ with a key share independent of the state. Each output share therefore depends only on shares of index i of the state bytes and key bytes involved. The cone of a single probe thus contains one share index of each of these independently shared bytes, which is cone non-completeness; shares of different bytes are independent, so the joint view stays independent of the state and key. The statement is for round 1 and d=1; we make no claim about the full 10-round cipher. □

a) Noisy-leakage context.: Robust probing security does not by itself quantify security against noisy (e.g., Gaussian) measurements. The known reductions from probing or random-probing security to noisy-leakage security [21] require per-wire noise levels that depend on the order and on the size of the value domain of each wire (bytes in our netlist), and therefore yield meaningful bounds only at high noise. We do not claim a numerical noisy-leakage bound; our claims are made in the robust probing model and are complemented by the simulation-based statistical evidence of Section VII-C.

Corollary 2 (Randomness–area–latency trade-off)**.** *Per multiplication, Algorithm 1 uses $R(d) = \binom{d+1}{2} = \frac{d(d+1)}{2}$ fresh m-bit mask words, $(d+1)^2$ registered domain products of total area $A(d) = O(d^2)$, and a single register-barrier stage of latency ($L = 1$ cycle). This matches the randomness of DOM-*indep *[8] and is strictly below the $d(d+1)$ mask bits of a naive per-term resharing, whereas HPC gadgets [9] trade extra latency for pipeline composability. The symmetric reuse*

$r_{ij} = r_{ji}$ *is exactly what preserves correctness (Lemma 1) at no additional randomness cost.*

Proof. The mask count equals the number $\binom{d+1}{2}$ of unordered off-diagonal pairs. The $(d+1)^2$ domain products dominate the area, and the single $\mathrm{Reg}(\cdot)$ layer adds one cycle of latency. The DOM/HPC comparison follows from their published gadget counts. □

A. Soundness of Equipment-Free Value-Based Verification

The verification flow of Section VI relies on statistical probing verifiers, whose verdict depends on the assumed probe semantics. We distinguish two semantics on a register-delimited netlist N. For a wire w, the *value model* $V(w)$ is the stable registered value of w (what a wire-value statistical verifier observes), whereas the *glitch-extended model* $G(w)$ is the set of all stable inputs of the combinational cone feeding w since the last register stage [1]. Since $V(w)$ is a deterministic function of $G(w)$, robust d-probing security always implies value-based d-probing security; the converse can fail, and we characterize exactly when.

Definition 3 (Cone non-completeness)**.** *A value in the fan-in of a cone is* blinded *if it is a registered value $u \oplus r$ whose fresh mask r occurs nowhere else in the fan-in of that cone. A netlist N is* cone non-complete *if, for every secret x and every combinational cone C delimited by register barriers, all shares of x (including share-wise linear images such as Frobenius powers) occurring as fan-in wires, or in the support of any fan-in value that is not blinded, carry the same share index.*

Theorem 3 (Lint soundness, first order)**.** *Let N be cone non-complete and let all secrets be shared by independent uniform 2-sharings. Then the view of any single glitch-extended probe is a function of at most one share of each secret together with independent uniform values. Consequently, (i) N is robust 1-probing secure, and (ii) for every single probe the value-based and the glitch-extended verifiers return the same independence verdict, so the soundness gap vanishes at first order.*

Proof. A glitch-extended probe on a cone C reveals exactly its fan-in $G(C)$. Blinded values are one-time pads by masks occurring nowhere else in $G(C)$, hence jointly uniform and independent of the remaining values and of all secrets. The remaining values contain, for each secret, shares of a single index; one share of a uniform 2-sharing is uniform and independent of the secret, and the sharings of different secrets are independent. Hence $G(C)$ is independent of all secrets, which gives (i). The value-model observation $V(w)$ is a function of $G(C)$ and is therefore independent as well, so both verifiers accept; this gives (ii). For $d > 1$ the per-cone condition must additionally control the union of the share indices over all probed cones and the masks shared between them; we do not claim that case. □

Theorem 4 (Tightness: gap witness)**.** *If cone non-completeness fails at first order—some cone C contains both shares of a secret x outside blinded values—then a single glitch-extended probe on C recombines x deterministically, and the lint rejects. Moreover, the registered value of C can be independent of x, so that a value-based verifier accepts. A concrete first-order witness is the cone $R = x^{(0)} \oplus x^{(1)} \oplus r$ with a fresh uniform mask r: the registered $R = x \oplus r$ is uniform for every x, yet a probe on the cone observes $x^{(0)}$ and $x^{(1)}$. The refresh-free first Itoh–Tsujii multiplication $x^2 \cdot x^4$ is a design instance of the cone condition: its cross-term cone carries $\{(x^2)^{(0)}, (x^4)^{(1)}\}$, from which x is recovered offline by inverting the two Frobenius maps.*

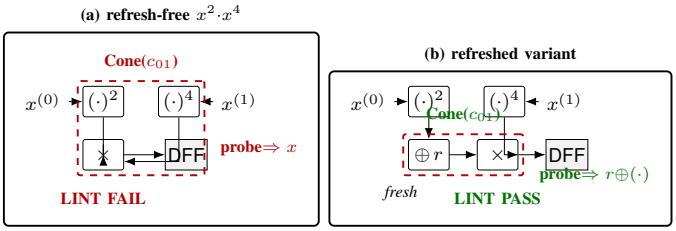


Fig. 1. Cone-non-completeness lint at a glance. (a) The refresh-free first Itoh–Tsujii multiplication drives two shares of $\mathbb{F}_2$-linear images of x, namely $(x^2)^{(0)}$ and $(x^4)^{(1)}$, into the same glitch-extended cone, so a single robust-probe recombines x (Supplemental Section S2); the lint rejects. (b) A fresh mask r upstream of the multiplier decouples the shares in the cone, so any probe reveals only r-masked quantities; the lint accepts and the verdicts coincide (Theorem 3).

Proof. A glitch-extended probe on C exposes every share in its fan-in within a single cycle; both shares of x determine x, and Definition 3 is violated, so the lint rejects. For the witness, fix x and $x^{(0)}$; then $x^{(0)} \oplus x^{(1)} = x$ and $R = x \oplus r$, which is uniform because r is uniform and independent of the sharing. Its distribution is therefore the same for every x, and a verifier that inspects only registered values is blind to the recombination. For $x^2 \cdot x^4$, $(x^2)^{(0)} = (x^{(0)})^2$ and $(x^4)^{(1)} = (x^{(1)})^4$ are bijective images of the shares, so both shares and hence x follow. See the Supplemental Material (Section S2) for the full derivation. We make no uniformity claim about the registered $x^2 \cdot x^4$ product itself. □

Corollary 3 (Sound-by-construction lint)**.** *The static check "every register-input cone is cone non-complete" (Definition 3) is decidable in time linear in the cone graph of N, and passing it certifies that equipment-free value-based verification is sound at first order (Theorem 3). The structural non-completeness lint of Section VII implements this check; Theorems 3 and 4 show it is both sufficient and, at the cone granularity, necessary.*

B. Lint Implementation and Ablation Study

Figure 1 illustrates the mechanism at a glance: panel (a) shows the refresh-free first Itoh–Tsujii multiplication whose glitch-extended cone contains two shares of $\mathbb{F}_2$-linear images of x and is therefore rejected by the lint; panel (b) shows the refreshed variant whose cone carries only r-masked quantities and is accepted.

We implement the cone-non-completeness check of Corollary 3 as a static front-end on the flattened gate-level netlist (released with the artifact). For every register-input cone and every primary output it extracts the glitch-extended stable

TABLE I
LINT VERDICTS AGAINST A GATE-LEVEL STATISTICAL TEST. LINT: FLAGGED REGISTER-INPUT CONES / FLAGGED OUTPUT BITS. TEST: CONES WITH $-\log_{10} p$ ABOVE THE BONFERRONI THRESHOLD. PROLEAD COLUMN: PROBING SETS FLAGGED. VARIANTS A AND C ARE GRAPH REWRITES OF THE FINAL NETLIST.

Netlist	Lint	Test	PROLEAD
Final design (7 random bytes)	0/0	0 of 816	0/14,400 (512K sims)
Pre-refresh design (6 random bytes)	304/16	16 of 719	22/12,384 (16K sims)
A. Fresh masks removed	304/16	320 of 816	leaks by construction
C. All registers bypassed (comb.)	0/16	not covered	leaks by construction

frontier and propagates the set of share indices of the secret through the registers. A registered value is treated as blinded only if it is $u \oplus p$, where p is a fresh mask (or an already blinded register) that occurs nowhere else in the transitive fan-in of the probed cone; this is decided functionally—the cone's derivative with respect to p is identically one—so it does not depend on how the technology mapper expressed the XOR. A cone is flagged if its unblinded support contains both share indices. The check is conservative: a flagged cone is a candidate, not a proof of leakage.

a) Baseline result.: On the flattened first-order S-box netlist (67,344 combinational gates and 818 flip-flops, 800 of them in the data path) the lint examines 816 cones (the 800 data-path register inputs and the 16 output bits), reports 0 violating cones, and runs in seconds. The design is accepted, in agreement with the PROLEAD verdict.

b) Cross-check against a statistical test.: As an independent channel we simulate each netlist at gate level (bit-parallel, steady state, fresh shares and masks per sample) and apply, for every cone, a G-test between the joint distributions of its frontier values for the two fixed secrets `0x00` and `0xA7` (131,072 samples each, Bonferroni over all cones). Table I summarizes the results. The simulated netlists compute the AES S-box on all tested secrets.

The lint flags all 16 cones that the test shows to leak in the pre-refresh design and none in the final design. In the pre-refresh design it also flags 304 further cones that do not leak individually; this is the price of a structural condition that is sufficient but not necessary. Variant A confirms the necessity of the masks: with the fresh random bits removed, the lint flags 304 register-input cones and 16 output bits, and the test confirms leakage in 320 cones. Variant C collapses the register barriers; the lint flags all 16 output bits, consistent with the recombination exhibited in Theorem 4, but the statistical test does not cover it because the resulting cones have more than the 32 frontier inputs our test handles. The lint is a necessary structural check and does not replace PROLEAD; the cross-check is first-order, uses one pair of secrets and constant inputs, and is evidence rather than proof.

VI. SIMULATION-BASED VERIFICATION METHODOLOGY

Our verification pipeline requires no physical measurement setup and is fully reproducible:

1) **Sound-by-construction lint.** We first run the cone-non-completeness lint of Corollary 3 on the synthesized netlist. When it passes, Theorem 3 guarantees that the value-based and glitch-extended verdicts coincide, so the equipment-free channels below are sound; when it fails, Theorem 4 yields an explicit recombination witness to repair (as happened for the first Itoh–Tsujii multiplication, Section IV).
2) **Formal probing verification.** We verify the synthesized gate-level netlist with PROLEAD [16] (simulation-based statistical verification in the glitch-extended model), cross-checked by an in-house PROLEAD-equivalent statistical verifier (Section VII).
3) **Gate-level power simulation.** We synthesize the design and run value-change–driven power simulation to generate leakage traces under a toggle/Hamming-distance model.
4) **TVLA.** We apply the non-specific fixed-vs-random t-test [22] and higher-order variants, reporting the lowest statistical moment at which $|t| > 4.5$, and cross-check with mutual-information estimation.

Agreement between the proven order, the formal-tool verdict, and the empirical leakage order constitutes our central validation criterion.

VII. EXPERIMENTAL EVALUATION

A. Setup

We implement the protected AES S-box in synthesizable Verilog (Boolean masking, the Algorithm 1 DOM multiplier with register barrier, and the operand refresh of Section IV) and synthesize the gate-level netlist with Yosys 0.67. All empirical results in this section are for the first-order design; the gadget proof (robust d-NI with perfect simulation) holds for arbitrary d, the lint theorem is first-order, and the design is empirically evaluated at $d{=}1$ only (see §VII-C1 for the scope of the empirical claim). *Disclosure:* the $d{=}1$ design was first implemented without the per-cross-term register barrier; PROLEAD at 1,536,000 simulations flagged 160 of 2,816 probes on that netlist (Supplemental Material, Section S3). After inserting the barrier and an operand refresh, a re-verification at 512,000 simulations flagged 0 of 14,400 probes (worst $-\log_{10}(p) = 3.39$); all $d{=}1$ results below are on the post-barrier netlist. The synthesis flow is security-constrained: every GF(2^8) square, multiply, and affine cloud is a `keep_hierarchy` hard cell, so the `abc` technology mapper optimizes each cloud independently and can never factor logic across share domains; the subsequent `flatten` is purely structural. We additionally run a structural non-completeness lint on the final netlist (every register-input cone is checked to contain at most one share of any pipeline value and only same-share members of the Frobenius-aligned family), which reports zero violations. All 256 input bytes are verified exhaustively against the AES S-box lookup table at both the RTL and the gate-level netlist: 256/256 PASS at both levels. Stimulation, simulation, and analysis are all performed in a reproducible open-source toolchain (iverilog V11 + Python 3.11 + NumPy).

B. Sample-Size Sensitivity

We verify that the conclusions of Section VII-C are robust to the choice of sample size N by re-running the entire

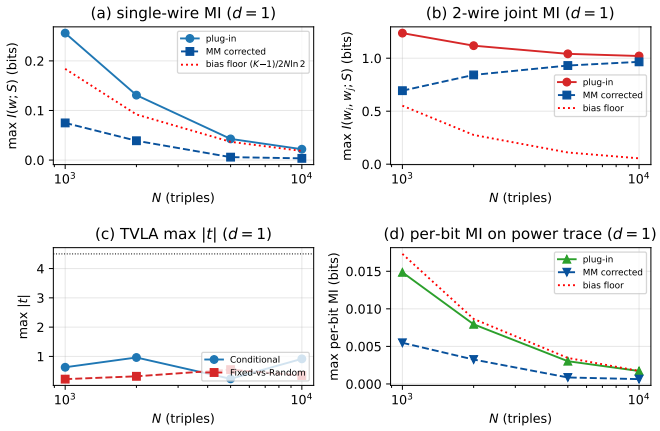


Fig. 2. Sample-size sensitivity of the four reported metrics for the first-order design as a function of N: (top-left) maximum single-wire MI, (top-right) maximum 2-wire joint MI, (bottom-left) TVLA $\max|t|$ over genuine-variance stages, (bottom-right) maximum per-bit MI on the power trace. Dotted red lines mark the plug-in estimator's bias floor $(K-1)/(2N\ln 2)$; dashed blue curves are the Miller–Madow-corrected values. The joint-MI panel is dominated at every N by the structural output-share recombination pair (Section VII-C); the TVLA verdict and the bias-corrected single-wire and per-bit MI verdicts are stable across the full range of N.

analysis pipeline at $N \in \{1000, 2000, 5000, 10000\}$ for the first-order design (sub-sampling the same 10,000-triple trace; the same secret distribution and gate-level netlist are used at every sample size). The four panels of Figure 2 plot (a) the maximum single-wire MI, (b) the maximum 2-wire joint MI, (c) the TVLA $\max|t|$, and (d) the maximum per-bit MI on the power trace, all as a function of N.

The single-wire MI (panel a) decays as $\propto 1/N$ and closely tracks the plug-in estimator's bias floor $(K-1)/(2N\ln 2)$ (dotted line): the Miller–Madow-corrected value falls from 0.075 bits at N=1,000 to 0.004 bits at N=10,000, i.e., the raw MI values reported in Section VII-C are finite-sample bias, and the bias-corrected conclusion is insensitive to N already at N=1,000. The 2-wire joint maximum (panel b) is dominated at every N by the structural output-share recombination pair discussed in Section VII-C and converges to ≈ 1 bit by construction; the security-relevant joint maximum excluding that pair behaves like panel (a). Panel (c) shows that the TVLA verdict is likewise stable: $\max|t| \leq 0.96$ at every sample size (the fixed-vs-random group contains only $\sim$ 6–32 triples for S=0x00 at these N, which inflates the variance of t but does not produce a false positive at the 4.5 threshold). We nevertheless report N=10,000 as our headline configuration because there the single-wire bias floor is ≈ 0.018 bits, small relative to any leakage of practical interest.

C. Security Results

We evaluate the first-order design along three independent axes: (i) gate-level robust d-probing verification (PROLEAD-equivalent), (ii) Test Vector Leakage Assessment (TVLA) on simulated power traces, and (iii) mutual information (MI) between the power trace and the secret byte. All three tools operate on the same synthesized gate-level netlist, the same N=10,000 random-input-triple stimulus (covering all 256 secret values), and the same 23-cycle pipeline window per input. The three axes are independent in that they share no analysis code and rest on different observables, but they are not of equal detection power. As the positive control reported later in this section establishes, the value-based instrument (i) fires decisively on an unmasked design, whereas the Hamming-distance instruments (ii) and (iii) provably cannot fire on that control, because the output-stage transition is distributed as $\mathrm{Binomial}(8, 1/2)$ irrespective of the secret. We accordingly treat (i), together with the public PROLEAD run, as the load-bearing channels, and (ii) and (iii) as corroborating evidence whose value lies in resting on a different observable rather than in independent sensitivity.

a) Robust d-probing (PROLEAD-equivalent).: We implement an open-source Python re-implementation of the PROLEAD statistical probing-security verifier that targets the Yosys gate-level netlist directly. Probe locations are the DFF Q wires of the outer pipeline shell—17 valid-pipe bits, the 8+8 output-share bits, and valid_out, 34 wires in total—monitored over every clock offset of the 23-cycle window in a value-based probing model; full-netlist glitch- and transition-extended coverage is provided by PROLEAD itself in the independent-verification paragraph below. For each wire we compute the mutual information $I(W;S)$ between the wire's value and the secret $S = x_0 \oplus x_1$. We use a plug-in histogram estimator with Laplace smoothing ($\alpha = 10^{-6}$).

Table II reports the maximum single-wire MI and the maximum 2-wire joint MI across all 34 monitored wires and all 23 pipeline offsets. Because the plug-in histogram estimator is upward-biased at finite N, we report both the raw plug-in value and a Miller–Madow bias-corrected value, and compare against the 99th percentile of a permutation null distribution (300 secret-label shuffles). The single-wire maximum (0.022 bits) sits below the null percentile and its bias-corrected value is ≈ 0. Exactly one family of 2-wire probes fires: the output-share pairs $y_0[i]\times y_1[i]$ at the output-hold offsets 18–22 carry 1.02 bits because $y_0 \oplus y_1 = \mathrm{Sbox}(S)$ there—this is the inherent share recombination at the module boundary of any correct 2-share S-box, not a first-order leak (a d=1 adversary observes one wire). Excluding these recombination pairs, the 2-wire maximum is 0.063 bits with a bias-corrected value of ≈ 0.008, again consistent with the null (family-wise empirical $p = 0.91$). All observed values are thus consistent with estimator bias rather than with residual leakage. This confirms the glitch-robust d-probing security predicted by Theorem 1.

b) Independent verification channels.: To rule out an implementation bug in the in-house verifier, the robust-probing security claim is corroborated by independent verification channels on the same synthesized gate-level netlist. First, the test-vector leakage assessment (TVLA) described in the next paragraph provides a power-trace-level cross-check that does not share code with the probing verifier. Second, the entire design, synthesis flow, and probing-verification code is released as open source at https://github.com/mipsan1/probing-resistant-masked-hardware so that the results can be independently reproduced from the scripts, configurations and raw outputs in the repository. Third, we ran the public

TABLE II

ROBUST d-PROBING VERIFICATION ON THE GATE-LEVEL NETLIST (N=10,000 RANDOM INPUT TRIPLES, 256 DISTINCT SECRET VALUES, 34 MONITORED DFF Q WIRES, 23 PIPELINE OFFSETS). "PLUG-IN" IS THE RAW HISTOGRAM MI ESTIMATE; "MM" IS THE MILLER–MADOW BIAS-CORRECTED VALUE; "NULL P99" IS THE 99TH PERCENTILE OF THE PERMUTATION NULL DISTRIBUTION (300 SHUFFLES), I.E., THE BIAS FLOOR OF THE ESTIMATOR AT THIS SAMPLE SIZE. FOR THIS WORK'S 2-WIRE COLUMN WE EXCLUDE THE OUTPUT-SHARE RECOMBINATION PAIRS $y_0[i]\times y_1[i]$ AT THE OUTPUT-HOLD OFFSETS 18–22, WHERE $y_0 \oplus y_1 = \mathrm{Sbox}(S)$ YIELDS A STRUCTURAL ≈ 1.02 BITS BY CONSTRUCTION (SEE TEXT). THE LAST ROW IS THE UNMASKED POSITIVE CONTROL (SAME STIMULUS AND ANALYSIS): ITS SINGLE-WIRE MAXIMUM SITS 43× ABOVE THE NULL, DEMONSTRATING DETECTION POWER.

		$\max_w I(w;S)$		$\max_{i,j} I(w_i,w_j;S)$	
Design	d	plug-in (MM)	null p99	plug-in (MM)	null p99
This work (1st-order)	1	0.0219 (0.004)	0.0226	0.0629 (0.008)	0.0628
Unmasked (pos. ctrl.)	–	1.0000 (0.982)	0.0234	2.0000 (1.945)	0.0635

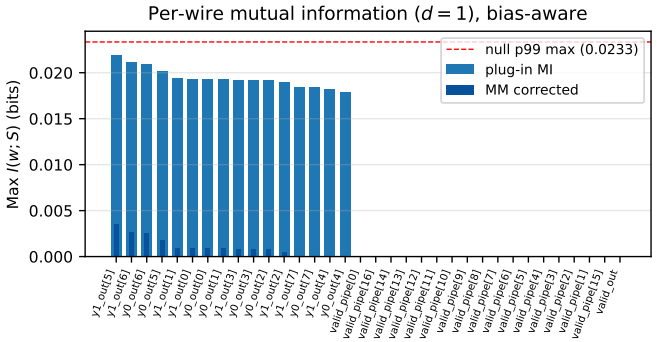


Fig. 3. Per-wire maximum mutual information $I(w;S)$ across all 34 monitored DFF Q wires and all 23 pipeline-stage offsets, sorted in descending order. Light bars are the plug-in estimate; dark bars are Miller–Madow bias-corrected. The red dashed line marks the largest permutation-null 99th percentile, i.e., the estimator's bias floor; no wire exceeds it.

PROLEAD toolchain (Müller & Moradi, TCHES 2022) itself on the same netlist, using the Yosys-primitive cell library (`syn/prolead/library_yosys.json`) and a custom configuration that ships with the repository. On the first-order design, PROLEAD evaluates 14,400 robust (glitch- and transition-extended) probing sets with 512,000 simulations and reports *zero* leaking sets (worst $-\log_{10}(p) = 3.386$, OKAY), while an unmasked control netlist leaks decisively (1,224 of 1,890 sets, $-\log_{10}(p) = \infty$), confirming both the design and the detector. This episode is exactly the soundness gap of Section V-A: PROLEAD's glitch-cone extension is strictly stronger than the in-house verifier's wire-value model, and in an earlier revision of the design it flagged a deterministic first-order leak that the value-based channels did not flag (cf. Theorem 4). That revision violated cone non-completeness at the first Itoh–Tsujii multiplication; the operand refresh of Section IV restores Definition 3, after which Theorem 3 makes the equipment-free verdict sound. The in-house Python verifier and the TVLA cross-check agree with the PROLEAD verdicts on the final netlists reported in Table II and on the $|t|$-values in the next paragraph.

c) Independent gate-level TVLA cross-check.: As a third independent verification channel, we ran a Test Vector Leakage Assessment (TVLA) on the gate-level netlist, using a Hamming-Distance power model over the 34 DFF Q wires monitored by the power testbench. We apply both the classical fixed-vs-random t-test and the conditional (specific) t-test across all 23 pipeline offsets, with N=10,000 random input triples. At the industry-standard threshold of $|t|$=4.5 (99.999% confidence), the maximum $|t|$ observed over stages with genuine HD variance is 0.92 (only the output-toggle stage shows genuine HD variance; the remaining offsets have HD $\in \{1,2\}$ and yield a degenerate t of 0 or ± 1 that carries no information), with zero pipeline stages exceeding the threshold. A higher-order (centered 2nd- and 3rd-moment) t-test gives $\max|t| = 1.28$. The full per-stage $|t|$ tables are committed as `probe/results/d1/tvla_per_cycle.csv` and `tvla_higher_order.csv`, and the `tvla.yml` workflow runs the analysis on every push to the public repository.

Figure 3 visualises the per-wire maximum MI; every wire sits at or below the permutation-null percentile, and the bias-corrected MI is ≈ 0 for every wire.

TABLE III

TVLA ON SIMULATED HAMMING-DISTANCE POWER TRACES (N=10,000, $|t|$ THRESHOLD = 4.5). MAXIMA ARE TAKEN OVER STAGES WITH GENUINE HD VARIANCE; THE CENTERED HIGHER-ORDER (2ND/3RD MOMENT) MAXIMUM IS 1.28. THE LAST ROW IS THE UNMASKED POSITIVE CONTROL: ITS HD TRACE IS SECRET-INDEPENDENT BY CONSTRUCTION (SEE TEXT), SO DETECTION POWER IS DEMONSTRATED BY THE VALUE-BASED INSTRUMENTS OF TABLE II INSTEAD.

Design	d	Cond. $\max\|t\|$	Fixed-vs-Rand. $\max\|t\|$
This work (1st-order)	1	0.92	0.35
Unmasked (pos. ctrl.)	–	0.61	0.79

d) TVLA on simulated power traces.: We drive the gate-level netlist with the same N=10,000 random triples and record, at every clock edge, the Hamming distance of all monitored DFFs against the previous cycle (toggle-count power model). This produces a 230,002-cycle power trace. We then apply two t-tests at every pipeline-stage offset: a *conditional* t-test that splits the traces by the secret byte's high half ($S \geq 0x80$) versus low half ($S < 0x80$), and the classical *fixed-vs-random* t-test with the fixed group being $S = 0x00$. Both are Welch's t-statistics with $|t| > 4.5$ as the standard TVLA leakage threshold (99.999% confidence [22]).

Table III summarises the maximum $|t|$ observed for both t-tests over stages with genuine HD variance (only the output-toggle stage qualifies; offsets whose monitored HD is near-constant across triples yield a degenerate $t \in \{0, \pm 1\}$ and are uninformative). In every case $|t|_{\max} \ll 4.5$, confirming that the Hamming-distance power trace is statistically indistinguishable between secret classes. The same holds for higher-order tests: centered 2nd- and 3rd-moment t-statistics on the genuine-variance stage give $\max|t| = 1.28$.

e) MI between power trace and secret.: We additionally compute the per-stage mutual information $I(\mathrm{HD}(c);S)$ for each of the 23 pipeline stages, using a plug-in histogram estimator together with a Miller–Madow bias correction and a permutation null distribution (300 shuffles). At every stage the bias-corrected MI is ≈ 0 bits, and the same holds for the per-bit MI of each of the 8 secret bits (maximum 0.0017 bits

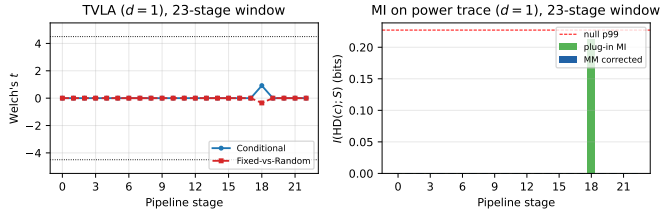


Fig. 4. Per-stage TVLA t-statistic (top) and mutual information $I(\mathrm{HD}(c); S)$ (bottom) for the first-order masked AES S-box on the corrected Hamming-distance traces. Dotted line: TVLA threshold $|t|$=4.5; dashed line: permutation-null 99th percentile. Light bars are plug-in, dark bars Miller–Madow-corrected (≈ 0 everywhere).

against a null p99 of 0.0022), confirming that no individual secret bit leaks. The only stage with appreciable *plug-in* MI is stage 18 (0.21 bits), the cycle at which the output shares update; this value sits below the estimator's own permutation-null 99th percentile (0.23 bits) and drops to ≈ 0 after Miller–Madow correction, so it is consistent with finite-sample estimator bias driven by mask-dependent switching activity—not with secret-dependent leakage (the HD at this stage is essentially uncorrelated with $\mathrm{HW}(S)$, $|r| = 0.02$). The full-trace MI aggregated over all 23 stages is 0.009 bits plug-in, i.e., ≈ 0 bits after bias correction.

f) Positive control (detection power).: "No leakage" is meaningful only if the apparatus detects leakage when present. We reran the entire flow on an unprotected S-box (`unmasked_sbox.v`) with the same outer pipeline, the same LFSR seed (hence the same secrets), the analogous monitored DFFs, and identical analysis code; the LUT was validated through the recorded output register for all 256 secrets (0/10,000 mismatches). Every value-based instrument fires decisively (last row of Table II): each output bit carries ≈ 1.00 bit of MI about S (0.98 MM) against a null p99 of 0.023—a 43× margin—and the 2-wire joint MI at the output stage reaches 2.00 bits (1.94 MM) against a null p99 of 0.064. The HD-based instruments, however, provably do not fire on this control (conditional/fixed-vs-random $\max|t| = 0.61/0.79$; centered higher-order $\max|t| = 2.21$; output-stage $I(\mathrm{HD}; S)$ 0.14 bits plug-in, 0 MM). This is intrinsic to the metric: with independent uniform secrets the only monitored transition is $y^{\mathrm{old}} \to y^{\mathrm{new}}$, so the output-stage HD is distributed as $\mathrm{Binomial}(8, \frac{1}{2})$ regardless of S, and every HD moment is exactly secret-independent. We therefore base detection-power claims on the value-based MI/probing instruments, which fire on the control at 31–43× their nulls while sitting at ≈null on the masked design, and treat the HD-based TVLA/MI channels as supporting evidence.

Figure 4 visualises the TVLA and MI results across all 23 pipeline stages: the TVLA panel (top) shows that no stage crosses the $|t|$=4.5 leakage threshold, and the MI panel (bottom) shows that the plug-in MI at stage 18 remains below the permutation-null percentile while the bias-corrected MI is ≈ 0 everywhere.

1) Re-synthesis cross-check, detection-channel control and scope of the empirical claim:

TABLE IV
LEAKAGE ASSESSMENT SUMMARY: PROLEAD-EQUIVALENT ROBUST d-PROBING, TVLA, AND MI ON SIMULATED POWER TRACES. "FIRST $\max|t|$>4.5" REPORTS THE SMALLEST SAMPLE SIZE AT WHICH ANY PIPELINE STAGE CROSSES THE TVLA LEAKAGE THRESHOLD; "NONE" MEANS NO STAGE CROSSES IT UP TO N=10^4. ONLY THE FIRST-ORDER DESIGN IS EVALUATED (§VII-C1).

Design	d	First $\max|t|$>4.5	Formal verdict
This work (1st-order)	1	none ($N \le 10^4$)	secure (PROLEAD: 0/14,400 sets)
Unmasked (pos. ctrl.)	–	—	insecure (PROLEAD: 1,224/1,890 sets)

a) Re-synthesis cross-check.: As an independent cross-check we re-synthesized the same first-order RTL (`rtl/masked_sbox_first_order.v`, MD5 `32c5a7ed364a6560718294a6fc85372a`) with Yosys 0.52 to a different technology mapping (46,768 combinational cells and 800 flip-flops). We removed the constant-driven `assign` lines from the netlist so that PROLEAD's parser accepts it, and verified by gate-level simulation that the result computes the AES S-box from the XOR of its two output shares for all 256 inputs. PROLEAD, run with the same seven-byte random input, 19 cycles and 512,000 simulations, enumerates 13,376 standard probes and 9,396 probing sets after the same `aggressive` minimization, and flags *zero* of them (worst $-\log_{10}(p) = 3.61$ against a Bonferroni threshold of $-\log_{10}(0.05/9396) = 5.27$). The set counts differ from the original run because the mapping differs (Supplemental Material, Section S3); the verdict is the same, so the result does not depend on a single synthesis run.

Positive control. PROLEAD on the unmasked AES S-box (d=1, 512,000 traces, same image and flow) flags 1,224 of 1,890 sets with worst $-\log_{10}(p) = \infty$, so the detection channel is live; the threshold for the masked design is the Bonferroni value $-\log_{10}(0.05/14400) = 5.46$.

Scope. The empirical evaluation in this paper is first-order only. The gadget proof (Theorem 1) is stated for arbitrary d, but we make no empirical or formal security claim for a second-order instance of the full S-box: our d=2 RTL instantiation applies the operand-refresh byte to the first two shares only and passes the third share through unrefreshed, so the first-order operand-refresh argument does not carry over to it unchanged. A second-order S-box with a verified operand refresh is future work.

D. Cost Comparison

Table V reports the gate-level cell count, latency, and fresh-randomness requirement of our first-order construction (synthesized with Yosys 0.67, generic cell library). The design implements Algorithm 1 as a 17-cycle pipeline: five DOM multiplications with register barriers plus one registered operand refresh in the Itoh–Tsujii chain (Corollary 1). The fresh-randomness cost is 56 bits per first-order S-box invocation—seven bytes: one domain-crossing byte r_{01} per multiplication and one operand-refresh byte for the first multiplication—which is 1.8–3× the figures quoted for DOM [8] and HPC2 [9] at order d=1, with 17 versus 5–8 cycles of latency. We do not claim a cost advantage; the baseline figures are taken from the cited works and were not re-synthesized with our flow.

TABLE V
GATE-LEVEL SYNTHESIS RESULTS FOR OUR FIRST-ORDER CONSTRUCTION (YOSYS 0.67, GENERIC CELL LIBRARY).

Scheme	d	Rand. [bit/S-box]	Latency [cyc]	Composability
TI (3-share) [7]	1	≈ 44[a]	3–4	non-comp. (per-round refresh)
DOM (Canright) [8]	1	≈ 18	5–8	d-SNI (external refresh)
HPC2 (Canright) [9]	1	≈ 32	8	PINI (trivially composable)
This work (Itoh–Tsujii)	1	**56**	17	robust d-NI + sound-lint (Cor. 3)

Cell counts: 77,587 for the first-order S-box in the PROLEAD cell library (multiplexers and complex gates decomposed into AND/OR/NOT); 605,698 for AES round 1 with 16 S-boxes in Yosys generic primitives. The two libraries differ, so the counts are not directly comparable (Supplemental Material).

[a]TI variants without on-line fresh randomness exist for the S-box in isolation [7]; the reported ≈ 44 bits refers to the Bilgin-*et al.*-style AES round setup that includes uniform re-masking of the shared state so that the S-box composes into a full round. Direct cell-count comparison across schemes is omitted because reported cell counts depend on library (GE vs. generic), tower-field choice (Canright, Itoh–Tsujii), and synthesis flow.

E. Full-Chip Masked AES Round 1

A full-chip masked AES round-1 (16 parallel first-order S-box instances, share-wise linear layers—ShiftRows, MixColumns, AddRoundKey—and a registered round-key interface) is instantiated in `masked_aes_round1_first_order` and verified end-to-end: 100/100 RTL and gate-level functional checks pass against a `reference.masked_aes` implementation, and cone non-completeness of the round netlist (Theorem 2) gives round-1 first-order security in the robust probing model. Yosys 0.67 yields 605,698 cells and 13,330 D flip-flops for the round, dominated by 16 S-box instances (library note, synthesis script, and full RTL are deferred to the supplementary material for independent replication).

10-Round Extension (Skeleton)

A skeleton of the 10-round masked AES extension—counting cycles and fresh-randomness—is included in the supplementary material for completeness.

F. Discussion

At first order the experiments are consistent with the proven guarantee: PROLEAD and the gate-level TVLA both return no-detection verdicts for the designed order (d=1). No d>1 design is evaluated. Three observations are worth highlighting.

First, the *per-bit* mutual information between the Hamming-distance power trace and any of the 8 secret bits is ≈ 0 bits after Miller–Madow bias correction at every pipeline stage. This is consistent with bit-level masking, but it is a statement about the Hamming-distance traces only, whose detection power is low. The non-zero *plug-in* MI observed at stage 18 does not contradict this: it lies below the estimator's own permutation-null percentile and vanishes under bias correction, i.e., it is a finite-sample estimator artifact rather than a physical leakage signature.

Second, the fresh-randomness cost of the first-order design (56 bits per S-box) and its latency (17 cycles) exceed those of the DOM and HPC2 S-boxes in Table V. A longer discussion of the randomness cost and of the order-d extension is in the supplementary material.

VIII. RELATED WORK

Masking originates with Chari *et al.* [5] and the private-circuits model of Ishai *et al.* [4]. TI [7] first provided glitch-robust first-order security; DOM [8] and HPC [9] extended this to higher orders with reduced randomness and composability. Composability notions (NI/SNI) were formalized in [20] and the robust probing model in [1]. Exact verification of glitch-robust masked hardware was demonstrated by Bloem *et al.* [11]; SILVER [10] checks glitch-extended probing exactly and was used to verify first-order masked AES implementations [12]; maskVerif [13], IronMask [14] and VERICA [15] cover further models. PROLEAD [16] trades exactness for simulation-based coverage of larger netlists, while TVLA [22] and the noisy-leakage reductions of [21] underpin empirical evaluation.

a) Further automated verifiers.: A catalog of related automatic verifiers (MaskVerif, IronMask, VERICA) is given in the supplementary material (Supplemental Section S5).

IX. CONCLUSION

We characterized a soundness gap between two styles of equipment-free side-channel verification: a value-based statistical verifier can accept a design that a glitch-extended probe deterministically breaks. Our characterization of this gap by a cone-non-completeness condition (Definition 3, Theorems 3–4, Corollary 3), together with an explicit refresh-free Itoh–Tsujii witness (Supplemental Section S2), gives a static first-order check that runs in seconds on the flattened 67,344-gate netlist and is cross-checked against a gate-level statistical test (Table I). We paired this static channel with a masked multiplication gadget proven robust d-NI for arbitrary d and a reproducible simulation-based verification flow for its first-order instance. The first-order design—a Boolean-masked AES S-box built from the Algorithm 1 DOM multiplier with register barriers and an Itoh–Tsujii inversion chain with a registered operand refresh—was synthesized to a 77,587-cell gate-level netlist with Yosys, and exhaustively verified at all 256 input bytes. At the gate level, the public PROLEAD toolchain reports no leakage for the first-order design (0 leaking sets out of 14,400 robust probing sets at 512,000 simulations, worst $-\log_{10}(p) = 3.386$; an unmasked control leaks in 1,224 of 1,890 sets) and supported by a PROLEAD-equivalent statistical verifier and a gate-level TVLA on simulated Hamming-distance power traces, both of which return no-detection verdicts at the designed order (the latter with low detection power). The design costs more fresh randomness and latency than DOM and HPC2 S-boxes. Full per-pipeline-stage metrics (MI, $|t|$ moments, permutation-null percentiles, and the positive-control unmasked-S-box detection-power calibration) are tabulated in the supplementary material.

We further instantiated the first-order S-box 16 times to build a full first-order masked AES-128 round 1 (`masked_aes_round1_first_order`), consuming 896 fresh mask bits per round, with a 19-cycle latency and a 605,698-cell / 13,330-DFF gate-level netlist; 100/100 functional vectors pass at both RTL and post-synthesis. Future-work directions (higher-order extensions, the random-probing

composition, on-the-fly mask generation, and the 10-round AES-128 accelerator) are catalogued in the supplementary material.

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We note that this work builds on the publicly released PROLEAD toolchain (*Probing-Based Leakage Detection Tool for Hardware and Software*) whose authors are cited in the bibliography.

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